NANYANG TECHNOLOGICAL UNIVERSITY
PH7002 MIDTERM EXAMINATION

Total: 100 points. Time allowed: 90 minutes

Oct 9th, 2025 (Thursday)

1. **Spins on triangle (30 points)**

   Assume three Ising spins are arranged in an equilateral triangle with each spin interacting with its two neighbors as shown in Fig. 1. The energy expression for the Ising model in a magnetic field is

$$H = -J(s_1 s_2 + s_1 s_3 + s_2 s_3) - \mu h(s_1 + s_2 + s_3).$$

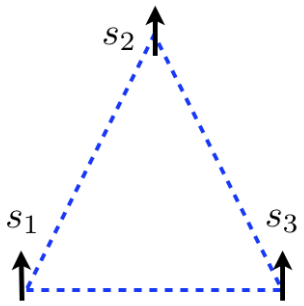


Figure 1: Ising spins on a triangle.

   Derive expressions for the partition function Z, the average spin $\langle s \rangle$ and the average energy $\langle E \rangle$.

2. **Landau theory (20 marks)**

   This problem deals with the Landau free energy for an Ising-like order parameter, m, in the absence of an external magnetic field. Suppose the free energy has the form

$$F(m) = \frac{a(T)}{2}\, m^2 + \frac{b}{6}\, m^4.$$

   Assume for this problem that the parameter a varies linearly with T and goes through zero at some temperature T_c, i.e. $a(T) = \alpha \cdot (T - T_c)$, and $\alpha, b > 0$ are temperature independent parameters. Sketch the free energy for $T < T_c$ and $T > T_c$. Find the equilibrium magnetization m for $T < T_c$ and $T > T_c$. What is the critical exponent β for $T < T_c$? Note: $m \sim (-t)^\beta$ where $t = (T - T_c)/T_c$. Is this a 1st or 2nd order phase transition?

3. **Biopolymer (50 points)**

Consider a simple model for biopolymer which is a linear chain of molecules. Let each molecule be in one of two states, (R) a right-handed loop or (L) a left-handed loop. If consecutive molecules are either both right-handed or both left-handed, there is a stabilizing interaction, with energy ($-J$). For a mismatched pair, the interaction energy is zero. A suitable Hamiltonian is the 2-state Potts model:

$$\mathcal{H} = -J \sum_{i=1}^{N} \delta_{s_i s_{i+1}}$$

where $s_i = +1$ (in R state) or -1 (in L state).

(a) Assume that the boundary conditions are unspecified and represented by the vectors $|a\rangle$ and $|b\rangle$ for the first and last molecule respectively. Write the partition function for a chain of N molecules. Show that the transfer matrix is

$$T = \begin{pmatrix} w & 1 \\ 1 & w \end{pmatrix}$$

where $w = e^{\beta J}$. Hence, evaluate the eigenvalues λ_+, λ_- (where $\lambda_+ > \lambda_-$) and the respective eigenvectors $|+\rangle$, $|-\rangle$ explicitly.

(b) Compute the free energy F. It suffices to find the extensive term proportional to N as $N \to \infty$.

(c) Compute $\langle R_n \rangle$, the probability that the n-th link is a right-handed loop R, for n in the middle of a long chain, i.e., $1 \ll n \ll N$.

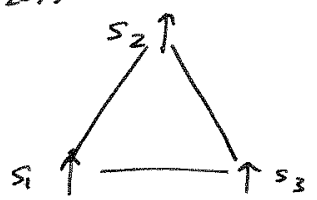
(d) Compute the correlation function

$$\langle R_n R_{n+m} \rangle$$

for a long chain, i.e., $1 \ll n \ll N$.

(e) At low temperature, $w \gg 1$, the correlation function becomes long-ranged. From the average R, what is the typical number of steps until we encounter a left-handed loop L?

$$H = -J(S_1S_2 + S_1S_3 + S_2S_3) - \mu h(S_1 + S_2 + S_3)$$



3)

$$Z = \sum_{S_1,S_2,S_3} e^{-\beta H}$$

(a)

Possible configurations

$$\left.\begin{array}{cccc} + & + & - & + \\ + \;\; + & - \;\; + & + \;\; + & + \;\; - \\ + & - & - & - \\ - \;\; - & - \;\; + & + \;\; - & - \;\; - \end{array}\right\} \text{8 in all}$$

let $K = \beta J$ and $P = \beta\mu h$

$$\Rightarrow \quad Z = e^{+3K+3P} + 3e^{-K+P} + 3e^{-K+P} + e^{3K-3P}$$

$$= 2e^{3K}\cosh 3P + 6e^{-K}\cosh P$$

(b)

$$\langle \sum_i S_i \rangle = +\frac{\partial}{\partial P}\log Z = \frac{6e^{3K}\sinh 3P + 6e^{-K}\sinh P}{Z}$$

$$\Rightarrow \quad \langle S \rangle = \frac{1}{N}\sum_i \langle S_i \rangle = \frac{2e^{3K}\sinh 3P + 2e^{-K}\sinh P}{Z}$$

$\boxed{N=3}$

(c)

$$\langle E \rangle = -\frac{\partial}{\partial \beta}\log Z = -\frac{\left(6e^{3K}J\cosh 3P - 6Je^{-K}\cosh P + \mu h\, 6e^{3K}\sinh 3P + \mu h\, 6e^{-K}\sinh P\right)}{Z}$$

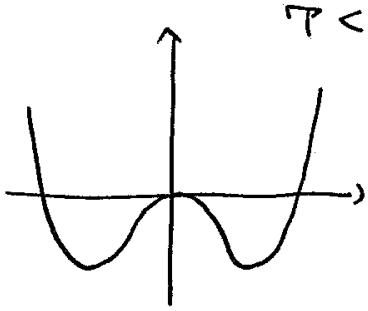
2) <u>Landau</u>

$$F(m) = \frac{a(T)}{2} m^2 + \frac{b}{4} m^4$$

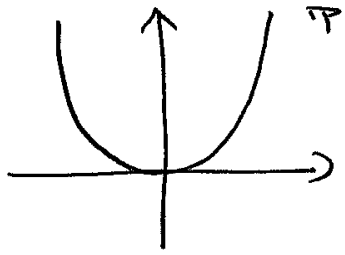
where $a(T) = \alpha(T - T_c)$

$\alpha, b > 0$

$T < T_c$



$T > T_c$



$\underbrace{T > T_c}_{a = \alpha(T - T_c) > 0}$ $\Rightarrow \frac{dF}{dm} = am + bm^3 = 0$

$m[a + bm^2] = 0$

$\Rightarrow m = 0$

$\underbrace{T < T_c}_{a < 0}$ $\Rightarrow \frac{dF}{dm} = am + bm^3 = 0$

$\Rightarrow m = 0$ or $\pm\sqrt{-\frac{a}{b}}$

$$m \simeq \pm\sqrt{-\frac{\alpha}{b}(T - T_c)} \sim |T - T_c|^{1/2}$$

$$\Rightarrow \beta = 1/2$$

2nd order phase transition.

This solution grows continuously from $m = 0$ at T_c.

a.) Let $s_i = \begin{cases} +1 & R \\ -1 & L \end{cases}$

$$\mathcal{H} = -J \sum_i \delta_{s_i, s_{i+1}}$$

$$Z = \sum_{s_i = \pm 1} \exp\left[\beta J \sum_i \delta(s_i, s_{i+1}) \right] = \sum_{s_i = \pm 1} (\omega)^{\delta(s_i, s_{i+1})}$$

b.) To compute the partition function, use the transfer matrix method

$$Z = v_L^T \, T^{N-1} \, v_R$$

where $T = \begin{pmatrix} \omega & 1 \\ 1 & \omega \end{pmatrix}$ and v_L, v_R depend on the boundary conditions. The eigenvectors of T are

$$v_+ = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} \qquad \lambda_+ = (\omega + 1)$$

$$v_- = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix} \qquad \lambda_- = (\omega - 1) \qquad \omega = e^{\beta J}$$

then $$Z = v_L^T v_+ \lambda_+^{N-1} v_+^T v_R + v_L^T v_- \lambda_-^{N-1} v_-^T v_R$$

for $N \gg 1$ $$\sim (\text{const}) \cdot \lambda_+^N$$

then $$\log Z = N \log \lambda_+ + O(1)$$

with $$e^{-\beta F} = \log Z \qquad F = -TN \log(e^{\beta J} + 1)$$

c.) $$\langle R_n \rangle = v_L^T T^n \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} T^{N-1-n} v_R / Z$$

[projection onto $s_i = R$] *

both T^n and T^{N-n} are dominated by v_+

$$\cong v_+^T \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} v_+ = \tfrac{1}{2}(1\,1)\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \tfrac{1}{2}$$

so $\langle R_n \rangle = \frac{1}{2}$ as expected

d.) $$\langle R_n R_{n+m} \rangle = v_L^T T^n \underline{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}} T^m \underline{\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}} T^{N-1-n-m} v_R / Z$$

$$\cong v_+^T \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} T^m \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} v_+$$

$$= \tfrac{1}{2}(1,1)\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\left[v_+ \lambda_+^m v_+^T + v_- \lambda_-^m v_-^T \right]\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix} / \lambda_+^m$$

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* Note: it is <u>not</u> right to put $R_n = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ and blindly copy the lecture notes.

so

$$\langle R_n R_{n+m} \rangle = \frac{1}{4}\left(1 + 1 \cdot \left(\frac{\partial_-}{\partial_+}\right)^m\right)$$

$$= \frac{1}{4}\left(1 + \left(\frac{\omega-1}{\omega+1}\right)^m\right)$$

or

$$\langle R_n R_{n+m} \rangle = \frac{1}{4}\left[1 + \left(\tanh \frac{\beta J}{2}\right)^m\right]$$

$$\longrightarrow \frac{1}{4} = \left(\frac{1}{2}\right)^2 \text{ as } m \to \infty$$

✓

e.) For $\omega \gg 1$ or $\beta \to \infty$

$$\frac{\omega-1}{\omega+1} \cong 1 - \frac{2}{\omega} = 1 - 2e^{-\beta J}$$

$$\left(\frac{\omega-1}{\omega+1}\right)^m = \exp\left[-2m e^{-\beta J}\right]$$

so the correlation length is $\xi = \left(\frac{1}{2} e^{+\beta J}\right)$

From at R, the probability that the next link is on T_1 is

$$\frac{1}{\omega} = e^{-\beta J}$$

This makes sense, I think,

$M = e^{\beta J}$ to see on T_1

$M' = \frac{1}{2} e^{\beta J}$ to destroy the correlation, on the average.